A16727W1

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B3. ATOMIC AND LASER PHYSICS

TRINITY TERM 2023

Friday, 16 June, 2:30 pm – 4.30 pm

Answer **two** *questions.*

Start the answer to each question in a **fresh book.**

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. (a) Explain what is meant by the LS-coupling scheme, stating the conditions under which it applies. In spectroscopic notation, give the terms that arise from the three configurations $3s^2$, $3s\,3p$ and $3s\,3d$. [5]

(b) Show that in the LS-coupling scheme the atomic g-factor is given by

$$g_J \simeq \frac{3}{2} + \frac{S(S+1) - L(L+1)}{2J(J+1)} .$$

[5]

(c) The ground electronic configuration of cadmium is $5s^2$. The light emitted from a cadmium lamp situated in a magnetic flux density of 1.5 T is observed (perpendicular to $\mathbf{B}$). The spectral lines are split into components with the frequencies listed in the table (given relative to $f_A = c/\lambda_A$ and similarly for f_B, f_C).

wavelength / nm	frequencies of components / GHz
$\lambda_A = 340.4$	$f_A, f_A \pm 10.5$
$\lambda_B = 346.8$	$f_B \pm 10.5, f_B \pm 21, f_B \pm 31.5$
$\lambda_C = 361.4$	$f_C, f_C \pm 10.5, f_C \pm 21, f_C \pm 31.5, f_C \pm 52.5$

All of these lines have the same upper term and the same lower term. Deduce the values of J and g_J for the levels involved in lines A and C. Give values of L and S for the two terms that are consistent with the values of g_J that you have deduced. [Note it is not possible to deduce which is the upper and which the lower term.] [8]

(d) Giving your reasoning, state what transition corresponds to line B. [3]

(e) List all the allowed transitions between these two terms (including those corresponding to lines A, B and C). For one of these lines (other than A, B or C) calculate the relative frequencies of its components, expressing your answer in the same form as in the table. [4]

2. (a) In the solution of the Schrödinger equation of a multi-electron atom, it is found that the radial wave function for the i'th electron can be written in terms of a function $P_{n,l}(r_i)$ that is a solution of the equation

$$\left[-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}r^2} + \frac{\hbar^2 l(l+1)}{2mr^2} + V(r)\right] P_{n,l}(r) = EP_{n,l}(r)\,.$$

Describe the approximations used to obtain this equation and sketch the form of the effective potential $V(r)$. [6]

(b) Rubidium (atomic number $Z = 37$) has a ground electronic configuration of 5s and an ionization energy of 4.177 eV. In the absorption spectrum of rubidium, lines are observed at the wavelengths:

794.8, 780.0, 421.6, 420.2, 359.2, 358.7 nm.

State the transitions that give rise to this series. Determine the values of the quantum defects δ_l for all the nl configurations involved in these transitions. Estimate the mean wavelength of the next doublet in this series. Comment briefly on why the pairs of lines get closer together along the series. [6]

(c) The quantum defect for the 4d electron is $\delta_{4d} = 1.23$. Explain why it is smaller than the values found in part (b). Determine whether the 4d configuration has a larger or smaller binding energy than that of 6s. [3]

(d) Laser light excites the transition at a wavelength of 421.6 nm and spontaneous emission is observed at the following wavelengths:

2791.3, 2293.9, 1528.8, 1475.2, 1366.5, 1323.5, 794.8, 780.0, 421.6 nm.

For each of these transitions, give the quantum numbers of the levels involved, explaining your reasoning. [6]

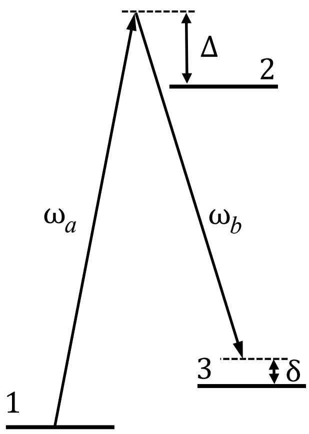
(e) Krypton ($Z = 36$) has an ionisation energy of 14.0 eV. What is the ground electronic configuration of krypton? Estimate the energy required to excite a krypton atom to an electronic configuration with a 5d electron, explaining your reasoning. [3]

3. In a two-level model of an atom the two eigenstates of the Hamiltonian $\hat{H}_0$ have energies E_1 and E_2 respectively ($E_2 > E_1$). This system is subjected to the time-dependent perturbation such that $\hat{H} = \hat{H}_0 + \hat{V}(t)$ where $\hat{V}(t) = ex\mathcal{E}\cos\omega t$ for an electric field of amplitude $\mathcal{E}$ along the x-direction oscillating at angular frequency ω. The general solution of the time-dependent Schrödinger equation can be written

$$\Psi(\mathbf{r},t) = c_1(t)\psi_1(\mathbf{r})\mathrm{e}^{-\mathrm{i}E_1 t/\hbar} + c_2(t)\psi_2(\mathbf{r})\mathrm{e}^{-\mathrm{i}E_2 t/\hbar} \ .$$

(a) Show that $\dot{c}_2(t) = -\mathrm{i}\Omega\cos(\omega t)\mathrm{e}^{\mathrm{i}\omega_{21}t}\, c_1(t)$, giving the expressions for ω_{21} and the Rabi frequency Ω. [6]

(b) Initially $c_2(0) = 0$ and the excitation is weak so that $c_1(t) \simeq 1$. Find $|c_2(t)|^2$, in the rotating-wave approximation. [6]



(c) Now consider a three-level model with eigenenergies E_1, E_2 and E_3 subjected to the time-dependent perturbation $\hat{V}(t) = ex\,(\mathcal{E}_a\cos\omega_a t + \mathcal{E}_b\cos\omega_b t)$, as illustrated in the figure (which is *not* to scale). The two frequency detunings $\Delta = \omega_a - \omega_{21}$ and $\delta = (\omega_a - \omega_b) - \omega_{31}$ are both small compared to all of the three angular frequencies: $\omega_{21} = (E_2 - E_1)/\hbar$, $\omega_{31} = (E_3 - E_1)/\hbar$ and $\omega_{23} = \omega_{21} - \omega_{31}$. For this system

$$\begin{aligned}\dot{c}_2(t) &\simeq -\mathrm{i}\Omega_a\cos(\omega_a t)\mathrm{e}^{\mathrm{i}\omega_{21}t}\, c_1(t)\\ \dot{c}_3(t) &\simeq -\mathrm{i}\Omega_b\cos(\omega_b t)\mathrm{e}^{-\mathrm{i}\omega_{23}t}\, c_2(t)\ .\end{aligned}$$

The initial conditions are $c_1(0) = 1$ and $c_2(0) = c_3(0) = 0$. The excitation is weak and $|\Delta| \gg |\delta|$. Show that

$$|c_3(t)|^2 = \frac{W^2}{\beta^2}\sin^2(\beta\, t)\ ,$$

justifying the approximations that you use and giving expressions for W and β. Comment on the comparison with the result for the one-photon transition in part (b). [8]

(d) A more general set of equations for $\dot{c}_2(t)$ and $\dot{c}_3(t)$ is

$$\begin{aligned}\dot{c}_2(t) &\simeq -\mathrm{i}\,(\Omega_a\cos(\omega_a t) + \Phi_b\cos(\omega_b t)\,)\,\mathrm{e}^{\mathrm{i}\omega_{21}t}\, c_1(t)\\ \dot{c}_3(t) &\simeq -\mathrm{i}\,(\Phi_a\cos(\omega_a t) + \Omega_b\cos(\omega_b t)\,)\,\mathrm{e}^{-\mathrm{i}\omega_{23}t}\, c_2(t)\ ,\end{aligned}$$

where $\Omega_a, \Omega_b, \Phi_a$ and Φ_b are Rabi frequencies of comparable magnitude. Describe why the simplified form of the equations in part (c) is a reasonable approximation. [5]

4. A homogeneously broadened transition of angular frequency ω_{21} occurs between two levels which have degeneracies g_1 and g_2. The upper level is populated at a rate R_2 per unit volume and decays only by spontaneous emission to the lower level, at a rate A_{21}. The lower level has a lifetime τ_1. The number densities for the lower and upper levels are N_1 and N_2 respectively.

(a) Write down the rate equations governing N_1 and N_2 and show that in the steady state the population inversion density $N^* = N_2 - (g_2/g_1)N_1$ is given by

$$N^* = R_2 \left(\frac{1}{A_{21}} - \frac{g_2}{g_1}\tau_1 \right) .$$ [4]

(b) Write down the rate equations governing N_1 and N_2 for this system in the presence of radiation of intensity I with a narrow bandwidth at angular frequency ω. Show that the gain coefficient at line centre has the form

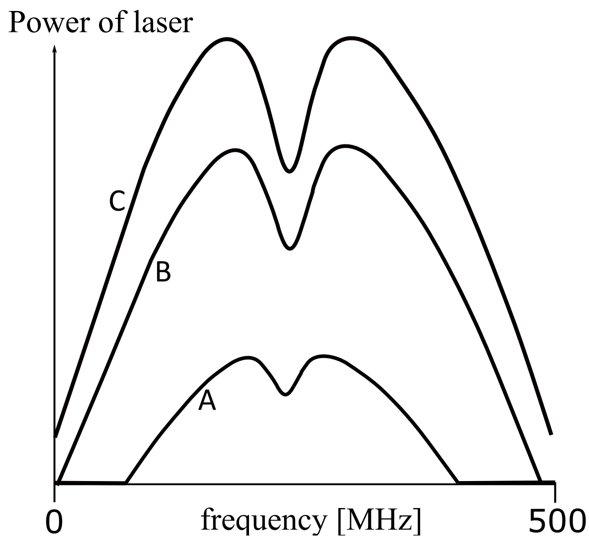
$$\alpha = \frac{\alpha_0}{1 + I/I_{\mathrm{sat}}} .$$

Give expressions for α_0 and I_{sat}. [6]

(c) Assuming a homogeneously broadened laser transition, sketch (all on the same scale) the frequency dependence of the gain coefficient experienced by the following narrow-band (e.g. laser) beams:
i) a weak beam,
ii) a beam that has an intensity equal to the value of I_{sat} at $\omega = \omega_{21}$ (the saturation intensity at line centre),
iii) a weak beam which overlaps spatially with a second laser beam which has a fixed frequency at the line centre ($\omega = \omega_{21}$) and an intensity I_{sat}. Explain how this case relates to those in i) and ii). [4]

This question is continued on the next page.

(d) In a helium-neon laser the gain medium is a gas mixture of ^{4}He and ^{20}Ne atoms that is excited by an electrical discharge. The laser cavity is formed by two mirrors at each end of the discharge tube. The figure shows the output power at a wavelength of 633 nm for three different values of R_2 (the rate at which the upper laser level of the neon atoms is selectively populated). The horizontal axis is the laser frequency. The dip in the gain curve has a Lorentzian line shape (inverted) with a width of 30 MHz (full width at half maximum) for curve A. State two examples of line-broadening mechanisms that are applicable to this gas-laser system. For each mechanism explain how its line width is related to physical properties of the system, and estimate values of these properties that are approximately consistent with the observed curves. [6]



(e) Explain why a dip in the gain appears at the line centre. The width of the dip increases with increasing excitation rate R_2 (corresponding to increasing the discharge current) and is 40 MHz (FWHM) for curve C in the figure. Suggest a mechanism that may cause this increase in the width. [5]